



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI BALAJI PETROLEUM CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

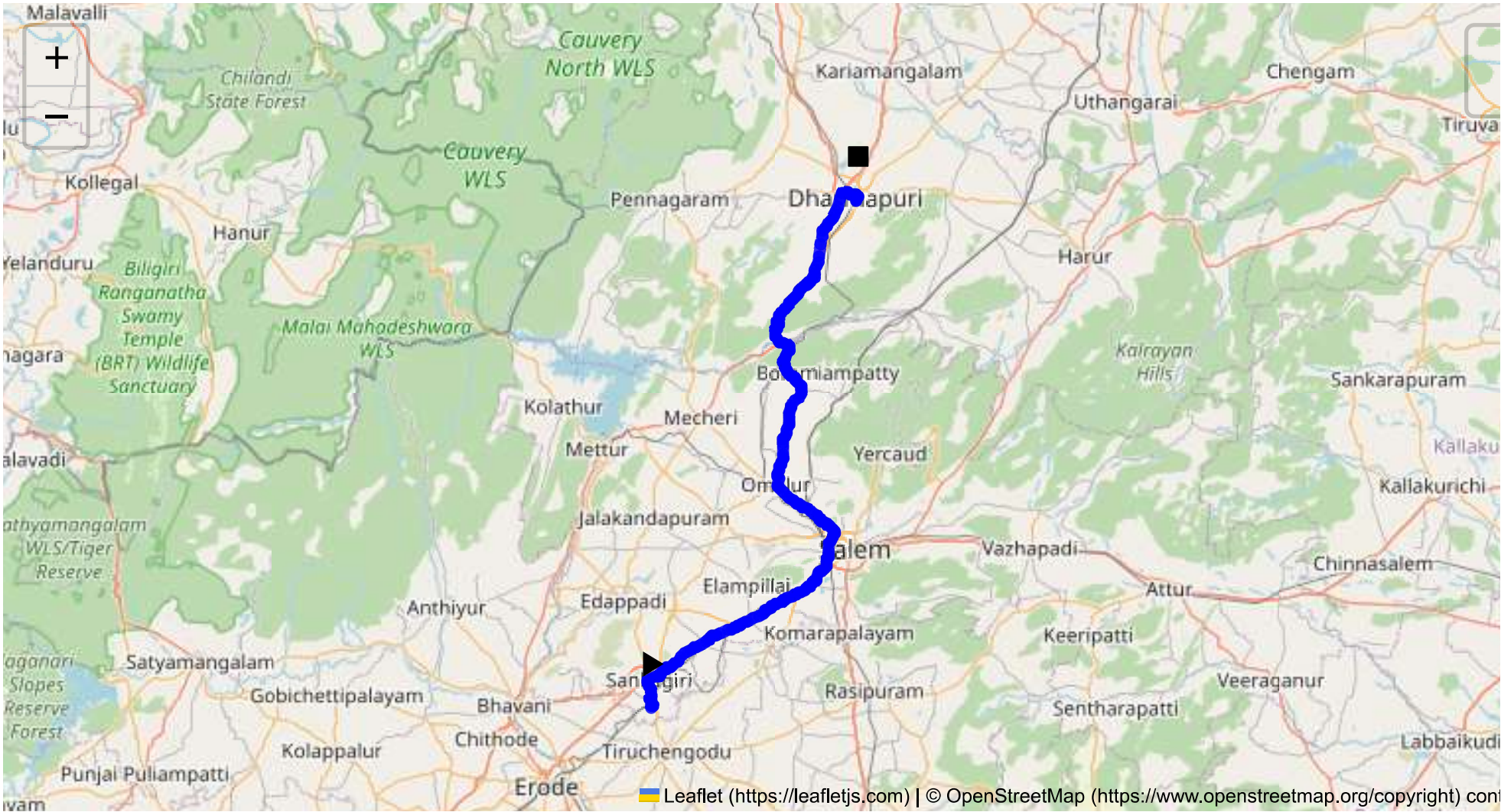
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 108.77 km
Estimated Duration: 2.1 hours
Adjusted Duration (Heavy Vehicle): 2.7 hours
Start: (11.4381, 77.8734)
End: (12.130415, 78.158605)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Dharmapuri, Tamil Nadu, India

1. Overview of the Route Map

The route from Sangagiri to Dharmapuri primarily follows the NH 544 from Sangagiri to Salem and then transitions to NH 44 towards Dharmapuri. The route spans approximately 108.77 kilometers, through a mix of urban, semi-urban, and rural landscapes.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences tropical weather. Key considerations include:

- **Monsoon Season (June to September):** Heavy rains can lead to waterlogging and reduced visibility.
- **Summer (March to May):** High temperatures can lead to road surface fatigue and possible overheating of vehicle engines.
- Periodic fog and mist in early mornings, especially in winter (December to February), may affect visibility.

3. Traffic Patterns

- **Peak Hours:** The stretch around Salem is especially congested during morning (8:00 - 10:00 AM) and evening (5:00 - 8:00 PM) peak hours.
- **Congestion-Prone Areas:** Salem city and junctions near bypass roads are main bottlenecks.

4. Assessment of Road Quality and Infrastructure

- **NH 544 and NH 44 are major highways** with generally good road conditions. However, look out for:
 - Occasional potholes post-monsoon.
 - Narrow segments in rural areas without median dividers.
- **Road infrastructure** is well-developed with regular toll plazas and rest areas.

5. Suggestions for Alternative Routes for Emergencies

- **Via SH 79:** This state highway can serve as an alternative during emergencies, connecting to parts of the route via regional roads with access to the main route at different points.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Adherence to specific lanes for heavy vehicles.
- Travel during designated times, often avoiding local city limits during peak hours to minimize risk.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Accidents are commonly associated with fatigue and overspeeding**, notably on curved segments and during adverse weather conditions.

- Few notable incidents of chemical spillages necessitate strict adherence to material handling protocols.

8. Environmental Considerations and Sensitive Areas

- Passing through villages necessitates caution to protect pedestrian and local animal movement.
- Agricultural lands adjacent to roads require careful driving to protect the environment.

9. Analysis of Communication Coverage

- **Generally good mobile network coverage** with occasional dead zones in rural patches, particularly in valleys or dense forested areas along NH 44.

10. Estimated Emergency Response Times for Different Route Segments

- Urban areas (Salem): 15-30 minutes from nearby stations.
- Rural stretches: Up to 45 minutes due to distance from emergency services.

12. Overall Summary of Risk Assessment

- **Moderate Risks** due to weather-induced hazards and traffic congestion.
- High-volume roadways reduce isolation but increase accident severity potential.
- Vital to monitor weather forecasts diligently and adhere to local transportation regulations.

Conclusion: This route, while primarily safe and well-maintained, needs careful navigation during harsh weather and busy periods. Adherence to regulations and proactive measures can mitigate most risks, ensuring a secure transit for hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.13658, 78.16171	15 KM/Hr
1	U-Turn	High	12.1302758, 78.15885469999999	10 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45000, 77.87213	30 KM/Hr
6	Turn	High	11.45028, 77.87202	15 KM/Hr
7	Turn	Medium	11.45309, 77.87325	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.46400, 77.87323	30 KM/Hr
9	Turn	Medium	11.46765, 77.87037	30 KM/Hr
10	Turn	High	11.47403, 77.86841	15 KM/Hr
11	Turn	High	11.49149, 77.89601	15 KM/Hr
12	Turn	High	11.49224, 77.89601	15 KM/Hr
13	Turn	Medium	11.53512, 77.95717	30 KM/Hr
14	Turn	Medium	11.53606, 77.95869	30 KM/Hr
15	Turn	Medium	11.64183, 78.11810	30 KM/Hr
16	Turn	Medium	11.64338, 78.11998	30 KM/Hr
17	Turn	Medium	11.95803, 78.04964	30 KM/Hr
18	Turn	Medium	11.95854, 78.04990	30 KM/Hr
19	Turn	Medium	11.96015, 78.04945	30 KM/Hr
20	Turn	Medium	11.96370, 78.05142	30 KM/Hr
21	Turn	Medium	11.96433, 78.05113	30 KM/Hr
22	Turn	High	11.96542, 78.04938	15 KM/Hr
23	Turn	High	11.96650, 78.04927	15 KM/Hr
24	Turn	Medium	11.96804, 78.05080	30 KM/Hr
25	Turn	High	12.14045, 78.13962	15 KM/Hr
26	Blind Spot	Blind Spot	12.13029, 78.15877	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
12	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
13	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
14	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
17	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
21	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
22	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
23	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
24	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
26	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
27	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
28	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
29	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
30	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
31	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
34	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
35	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
37	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
38	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
40	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
42	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
43	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
44	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
46	hospital	Vijaya Hospital	12.1399758, 78.1532115	30 km/h	Medium
47	clinic	Bee Tee Clinic	12.138889, 78.1576748	30 km/h	Medium
48	clinic	Sheela Clinic	12.138725, 78.1573436	30 km/h	Medium
49	clinic	Suguna Clinic	12.1386752, 78.1575468	30 km/h	Medium
50	hospital	Navina Hospital	12.1367521, 78.1610296	30 km/h	Medium
51	clinic	Bharani Neurologist	12.135969, 78.16255	30 km/h	Medium
52	hospital	Pragadeesh Hospital	12.135284, 78.1591867	30 km/h	Medium
53	hospital	D.N.V Hospital	12.1351328, 78.162869	30 km/h	Medium
54	clinic	Ragavendra Physiotherapy Clinic	12.135569, 78.1635247	30 km/h	Medium
55	hospital	Kamatchi Hospital	12.1350608, 78.1594841	30 km/h	Medium
56	clinic	Malar Siddha Clinic	12.1350485, 78.159226	30 km/h	Medium
57	hospital	Kamala Nehru Hospital	12.13404, 78.1633476	30 km/h	Medium
58	clinic	Govindaraj Clinic	12.1344914, 78.1642372	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
59	hospital	Sri Priya Hospital	12.135347, 78.164391	30 km/h	Medium
60	hospital	Vasan Eye Care-Dharmapuri	12.1326054, 78.1630215	30 km/h	Medium
61	hospital	Supa Hospital	12.1325709, 78.1625215	30 km/h	Medium
62	hospital	K.S. Hospital	12.1324134, 78.1620136	30 km/h	Medium
63	hospital	K. B. Hospital	12.1327051, 78.1612833	30 km/h	Medium
64	clinic	Kamalam Nursing Home	12.1318108, 78.1589049	30 km/h	Medium
65	clinic	Om Sakthi Nursing Home	12.1324, 78.161576	30 km/h	Medium
66	hospital	The Light Eye Hospital	12.1321184, 78.1594	30 km/h	Medium
67	hospital	Kumudha Hospital	12.1324596, 78.1620599	30 km/h	Medium
68	hospital	Gowri Eye Hospital	12.1327566, 78.1618526	30 km/h	Medium
69	hospital	Mahendran Dental Hospital	12.1328544, 78.1615842	30 km/h	Medium
70	clinic	Maruthi Mind Care	12.1335944, 78.1592428	30 km/h	Medium
71	clinic	Sugam Polyclinic	12.1323296, 78.1629319	30 km/h	Medium
72	hospital	Raaj Ortho Care Hospital	12.1325581, 78.163019	30 km/h	Medium
73	hospital	P.M.K. Hospital	12.131546, 78.1619069	30 km/h	Medium
74	clinic	ABC Clinics	12.129608, 78.157027	30 km/h	Medium
75	hospital	BGR Hospital	12.1308406, 78.1610398	30 km/h	Medium
76	hospital	Venkatesh Ortho Trauma Hospital	12.1301667, 78.1614714	30 km/h	Medium
77	hospital	SSS Hospital	12.1313019, 78.1603435	30 km/h	Medium
78	hospital	Sri Annai Hospital	12.1299576, 78.1614455	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
79	hospital	Arul Hospital	12.1309667, 78.157854	30 km/h	Medium
80	hospital	Rajarajan Hospital	12.1308139, 78.1614702	30 km/h	Medium
81	clinic	Med Health Superspeciality Clinic	12.1312842, 78.1592494	30 km/h	Medium
82	hospital	K. J. Hospital	12.1281634, 78.1579522	30 km/h	Medium
83	hospital	Mary Hospital	12.1283666, 78.1601592	30 km/h	Medium
84	hospital	Ap Hospital	12.128501, 78.157373	30 km/h	Medium
85	clinic	Vinod Child Care	12.1281934, 78.1578389	30 km/h	Medium
86	hospital	MMM Children & Skin Care Hospital	12.1282265, 78.1576734	30 km/h	Medium
87	clinic	Guru Bone & Joint Care	12.127949, 78.157032	30 km/h	Medium
88	hospital	Samyuktha Hospital	12.1283621, 78.1606407	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
18	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
19	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
20	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium